Innovative Green Roofing and PV Integration for Energy-Positive Parking Structures

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Abstract— **This study details the concept and implementation of an innovative green roof car park fitted with an advanced solar tracking photovoltaic array. The system design prioritizes synergy between power generation, microclimate improvement, and visual integration. Tracking PV structures are adapted to the roof's green zones, maximizing solar input while minimizing vegetation shading through computational optimization. Suitable plant species, resilient to partial shade and fluctuating irradiance, are investigated and monitored over time. Empirical results reveal that, given optimized mounting and selection, growth rates and biodiversity are maintained, while panel energy production approaches theoretical maxima for dual-axis tracking systems. The findings demonstrate a successful model for harmonizing renewable energy infrastructure with urban ecology, offering actionable guidance for future green/solar co-deployments in parking environments.**

Keywords—green roof; car parks; solar tracking; solar photovoltaic

I. Introduction

In order to increase water infiltration to the ground and reduce surface temperature in car parks, permeable pavement solutions and shading structures are the most common measures [1]. Another interesting solution, which is still not frequent but becoming increasingly popular due to its economic and environmental benefits, is the implementation of shading systems with photovoltaic (PV) panels [2]. In this paper, an innovative solution consisting of a shading system which integrates green roofs modules and photovoltaic panels with sun tracking is explored. While sun tracking generally increases the initial cost of installation, it can lead to the production of more electricity (up to 30% more when compared with fixed panels [3]) and is particularly interesting when the available area is scarce or too costly. The long-term objective of this work is to develop a shading system that simultaneously contributes to the mitigation of the urban heat island effect, the prevention of flooding by reducing runoff water volume, the improvement of air and water quality and the reduction of noise pollution, while allowing for the production of electricity for vehicle charging, night lighting, etc., or to be injected in the grid directly. The complete system being proposed is composed by a support structure, a solar photovoltaic system, a solar tracker and a green roof. Fig. 1 shows the novel concept proposed.

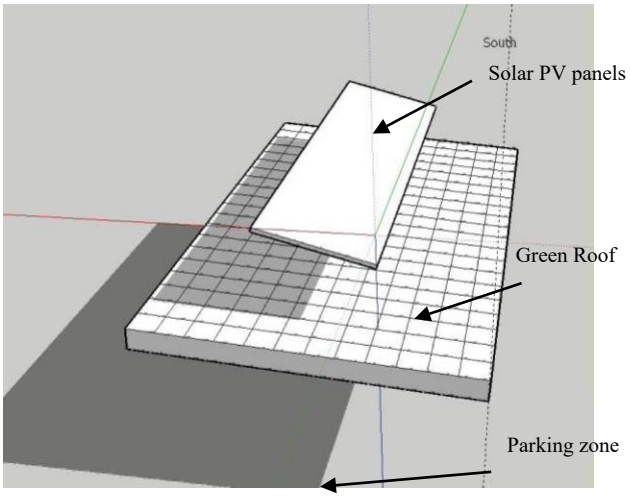


Fig. 1. Elements of Green Roof system for car parks proposed

In this paper, the emphasis is placed on the design of the solar tracking system and green roof and on their interaction. In the following sections, a dynamic model of the solar tracking is first developed to compute and compare the energy yield of different types of solar tracking. Afterwards various tracking options are studied and a selection is made based on the energy yield, geometric constraints, wind loads, and shadowing of the tracker. In the next section the development of the mechanical design of the actuator and tracking mechanism is presented. In the section 3 of this paper, a 3D optical model is developed to evaluate the interaction of the solar tracking radiation with the green roof. In the subsequent section, and based on the obtained results, the species of plants for the green roof are selected and results of its growth are presented. In the final section, the main conclusions of this study are drawn.

II. Solar Tracker

A. Selection

The design criteria for the solar tracker development was to optimize the production of energy throughout the year, while minimizing the aesthetical impact, wind load, and shadowing of the green roof. The different possibilities for tracking were analysed, regarding these multiple criteria, and the results are shown in Table I. TABLE I. Solar Tracker Selection

Table Head	Table Column Head		
	One-axis inclination	***One-axis azimuth***	***One-axis polar***
Energy gain [(a)]	24.6%	35.3%	36.8%
Integration	simple	difficult	difficult
Wind load	low	low	high
Aesthetical impact	low	low	high
Shadowing	low	high	low

[a.] Compared with fixed horizontal

Taking these multiple criteria into account, a one-axis system with inclination tracking was selected. This provides significant energy increase, good integration with the structure, a low wind load, good aesthetics, and reduced shadowing of the green roof.

B. Design

Once the type of solar tracking system was selected, the actuator design was developed. The design variables considered were: its position relative to the main structure, its range of movement and rotation angles. Taking into account a preliminary computation of the weight of the structure, the effect of the wind on the structure, as well as imposed loads defined in the RSA code [4], the maximum load obtained for a system with an actuator at either end of the assembly was 8 kN per actuator. Fig. 2 illustrates the actuator for an inclination of 30°.

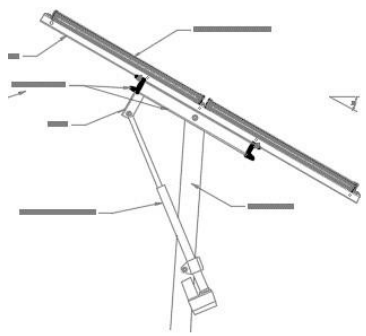


Fig.2 Solar tracking system design proposed

As shown in Fig. 2, the main support of the solar tracking structure is also tilted at a small angle. This small inclination was considered in order to provide higher stability when it is linked to rest of the car park structure since the green roof is also tilted with a small gradient for drainage purposes. The maximum angles of rotation of the system were selected taking in account multiple criteria, such as the energy yield, and interaction with the green roof. The computation of the energy gain dependence on maximum rotation angle was performed using the System Advisor Model software. Fig. 3 shows the results of the model simulations for various maximum angles of rotation.

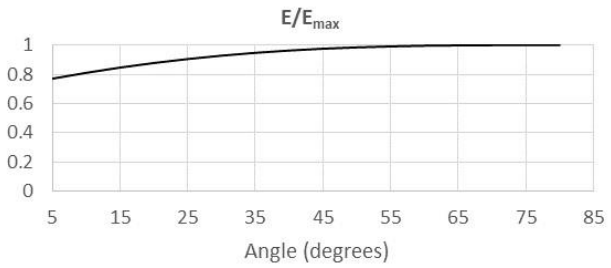


Fig.3 Solar tracking efficiency in function of maximum angle

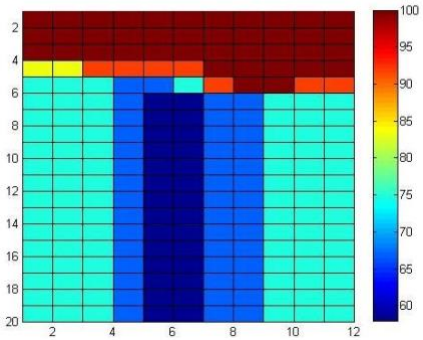
It can be observed that a maximum angle of 30 degrees allows the use of standard 8 kN actuators with a stroke of 600 mm, while reaching 93% of the maximum theoretical energy of that tracking mode, as well as providing greater sunlight exposure on the vegetation of the green roof.

III. Green Roof

A. Design

The design of the green roof was performed taking into consideration its exposure to extreme conditions such as elevated temperatures, shallow soil, wind and high light exposure. Green roofs are known to present harsh challenges for most plant species. For these reasons, plant selection requires the consideration of several key factors like microclimate, maintenance requirement, substrate depth and functional purposes of the green system project. Furthermore, when equipped with solar panels, green roofs provide additional challenging conditions such as shade and low access to rain water for part of the vegetation. Within the scope of the present study, a low maintenance and shallow substrate green roof vegetation blanket of pre-cultivated plants was selected. The mat was composed of a broad mix of plant species (33 herbaceous perennials and annuals, 4 stonecrops and 7 grass species) which provide a highly diverse community able to adapt to a wide range of green roof conditions. A preliminary study was carried out in order to assess the tolerance to the shading of the selected green roof blanket caused by the solar tracking system. In this assay, plants were cultivated during a whole vegetative cycle (from January to November) under different conditions of solar radiation distribution in order to simulate this effect.

An optical model was developed to analyse the 3D blockage of solar radiation caused by the tracking system and compute the availability of direct solar radiation on the different zones of the green roof for the different seasons of the year. To model the solar geometry and tracking angles the System Advisor Model (SAM) [6]. The two-dimensional radiation distribution on the mat was computed using Google Sketchup. The results were postprocessed and analysed in a MATLAB code. The computation of direct solar radiation was performed on an hourly basis discrete simulation for 4 reference days, representing different seasons. The results were timeintegrated for each spatial position in the mat to obtain the distribution of the radiation in each position. Fig. 4 shows the two-dimensional distribution of hours of direct solar radiation available on each position of the green roof for the different seasons of the year.



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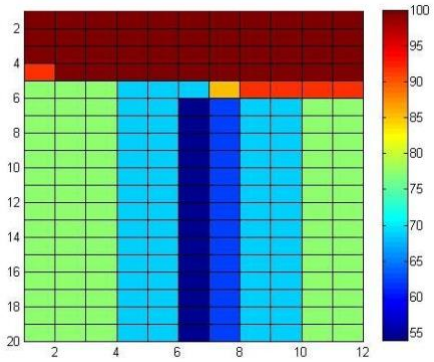

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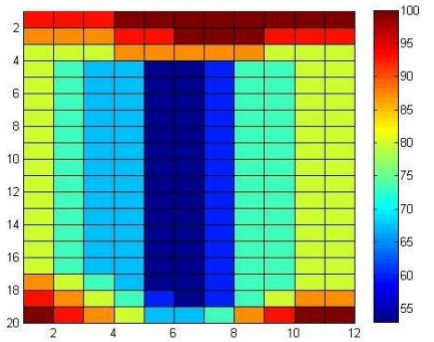

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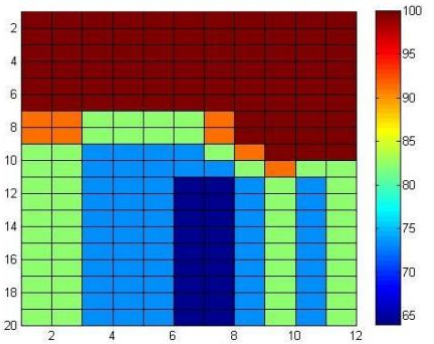

d)

Fig.4 Availability of hours of direct solar radiation on the green roof
a) 23 September b) 20 March c) 21 June d) 21 December

It can be concluded that the solar tracking configuration selected allows reaching a minimum of half-shading conditions for all the mat zones, in all the seasons.

B. Results

The results of plant growth and height, flowering, ground cover and blanket biodiversity were monitored comparing full and the partial shade conditions expected for the green roof. The survey did not display any striking differences between full sun and the lowest light exposure vegetation which suggests that the partial shade conditions did not represent a limiting factor to most plant species that compose the green roof mat.

Fig.5 Green roof species development results

IV. Conclusions

The integration of a one-axis solar tracker and green roof for a car parking system was developed and optimized. The results show good integration between the solar tracking mechanism and green roof allowing for an increase of 22.9% in the energy yield, when compared with horizontal photovoltaic systems, while maintain compatibility with the green roof by limitation of the rotation angle to 30 degrees. The findings also confirm the resistance of the selected mix of plant species to the partial shade exposure conditions that are expected for the green roof due to the interaction with the developed solar tracker.

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